10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 14 — Light

Solved Exercise | New Book (PECTAA 2026)

Exercise 14 (Solved)

This exercise covers reflection in mirrors, refraction and refractive index, Snell's law, total internal reflection and critical angle, optical fibres, lenses, the power of a lens and the defects of vision.

A. Multiple Choice Questions

14.1 What type of image is always formed by a convex mirror?

(a) Real and inverted
(b) Virtual and diminished
(c) Real and magnified
(d) Virtual and magnified

Answer: (b) Virtual and diminished — a convex mirror always forms an erect, virtual and smaller image behind the mirror, whatever the position of the object — that is why it is used as a rear-view mirror.

14.2 The optical device which uses a convex lens to magnify an object by creating a virtual image:

(a) camera
(b) microscope
(c) magnifying glass
(d) slide projector

Answer: (c) magnifying glass — with the object nearer than the focal length, a convex lens gives an erect, virtual and magnified image — this is the simple magnifier.

14.3 Which statement best describes the phenomenon of total internal reflection?

(a) Light bends when it enters a new medium
(b) Light is completely reflected at the boundary of two media
(c) Light slows down as it enters a denser medium
(d) Light passes through a medium without changing direction

Answer: (b) Light is completely reflected at the boundary of two media — when light travels from a denser to a rarer medium beyond the critical angle, the whole of it is reflected back into the denser medium.

14.4 A ray of light travelling from water to air bends:

(a) toward the normal
(b) away from the normal
(c) at 90° to the normal
(d) parallel to the normal

Answer: (b) away from the normal — light speeds up on going from water to air and bends away from the normal, because the angle of refraction is greater than the angle of incidence.

14.5 The optical device which works on the principle of inverting an image for projection is:

(a) magnifying glass
(c) slide projector

studysapiens.com

(b) photograph Enlarger (d) plane mirror

Answer: (c) slide projector — in a slide projector the slide is placed beyond the focal length, so a convex lens forms a real, inverted and magnified image on the screen.

14.6 What happens to light when it enters an optical fibre?

(a) It is absorbed by the fibre (c) It gets refracted at every point
(b) It undergoes total internal reflection (d) It travels in a straight line without deviation

Answer: (b) It undergoes total internal reflection — the light strikes the core–cladding boundary at an angle greater than the critical angle again and again, so it is trapped inside the core and carried along the fibre.

14.7 Which of the following phenomenon explains why a pencil appears bent when partially immersed in water?

(a) Reflection (c) Refraction
(b) Diffraction (d) Interference

Answer: (c) Refraction — light from the part of the pencil under water bends on coming out into the air, so the pencil appears shifted and hence bent at the surface.

B. Short Answer Questions

14.1 What happens when light passes from air to water?

When light passes from air into water, it goes from a rarer medium into a denser medium, so its speed decreases — from 3×10^8 m s^{-1} in air to about 2.25×10^8 m s^{-1} in water.

Because of this change of speed the ray bends at the surface. If it enters at an angle, it bends towards the normal, and the angle of refraction is smaller than the angle of incidence. If it enters along the normal (angle of incidence 0°), it goes straight on without bending. The frequency of the light does not change, but its wavelength becomes shorter in water. This bending is called refraction, and it is why a coin in a pool looks raised and a pencil looks bent at the water surface.

14.2 Define refractive index. What is the refractive index of air and water?

The refractive index of a medium is the ratio of the speed of light in vacuum (or air) to the speed of light in that medium: n = c / v. It is a pure number with no unit and is always greater than 1 for a material medium.

It can also be written, by Snell's law, as n = sin i / sin r, when light goes from air into the medium.

For air the refractive index is about 1.00 (very nearly 1), and for water it is about 1.33. For glass it is about 1.5 and for diamond about 2.42. A larger refractive index means the light travels more slowly in that medium and bends more.

studysapiens.com

14.3 What is the principle behind optical fibres?

An optical fibre works on the principle of total internal reflection. It is a very thin, flexible thread of glass or plastic made of two parts: an inner core of high refractive index and an outer cladding of slightly lower refractive index.

Light entering one end of the fibre is travelling in the denser core. When it reaches the boundary with the rarer cladding, it strikes at an angle greater than the critical angle, so it is totally reflected back into the core instead of being refracted out. This happens again and again along the length of the fibre, so the light is trapped and carried from one end to the other, even if the fibre is bent.

Because the signal is reflected and not lost, optical fibres carry telephone calls, internet data and TV signals over very long distances with very little loss, and they are also used in endoscopes to look inside the body and in decorative lamps.

14.4 Describe briefly how a convex lens forms a real or virtual image.

A convex lens is thicker in the middle and converges the rays that pass through it. The kind of image it forms depends on where the object is placed.

Real image: when the object is placed beyond the focal point (at a distance greater than f), the refracted rays actually meet on the other side of the lens and form a real, inverted image. If the object is between f and 2f the image is real, inverted and magnified; if the object is beyond 2f the image is real, inverted and diminished. This is the arrangement used in a camera and in the eye.

Virtual image: when the object is placed within the focal length (distance less than f), the refracted rays diverge and only appear to meet on the same side as the object. The image so formed is virtual, erect and magnified, and it cannot be taken on a screen. This is how a magnifying glass works.

studysapiens.com

14.5 How does the power of a lens relate to its focal length?

The power of a lens is the measure of its ability to bend (converge or diverge) light. It is defined as the reciprocal of the focal length measured in metres: $P = 1 / f$.

The unit of power is the dioptre (D), where $1\ D = 1\ m^{-1}$. So a lens of focal length 0.5 m has power $P = 1 / 0.5 = 2$ D, and a lens of focal length 2 m has power 0.5 D.

Power is inversely proportional to focal length: a short focal length means a strong converging (or diverging) power, and a long focal length means weak power. The power of a convex (converging) lens is taken as positive and that of a concave (diverging) lens as negative. Opticians write spectacle prescriptions in dioptres for this reason.

14.6 Write Snell's law and define the terms used in the equation.

Snell's law states that when light passes from one transparent medium into another, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for the two media. This constant is the refractive index of the second medium with respect to the first.

$$n = \sin i / \sin r$$

$$\text{or } n_1 \sin i = n_2 \sin r$$

Terms: i is the angle of incidence, the angle between the incident ray and the normal at the point of incidence; r is the angle of refraction, the angle between the refracted ray and the normal; n (or n_2/n_1) is the refractive index, the ratio of the speed of light in the first medium to that in the second, $n = v_1 / v_2$.

The angles are always measured from the normal, not from the surface. When light goes from a rarer to a denser medium it bends towards the normal ($r < i$), and when it goes from a denser to a rarer medium it bends away from the normal ($r > i$).

C. Constructed Response Questions

14.1 When a pencil is partially dipped in water, it appears bent or broken. What principle of light explains this phenomenon, and how does it relate to changes in speed?

The principle is refraction of light — the bending of light when it passes from one transparent medium into another because of a change in its speed.

The part of the pencil under water is seen by light that travels from the pencil through water and then out into the air. Water is optically denser than air, so the light travels more slowly in water (about 2.25×10^8 m s^{-1}) than in air (3×10^8 m s^{-1}). At the surface the rays bend away from the normal while leaving the water. Our eyes, however, assume that the light has travelled in a straight line, so we see the underwater part of the pencil shifted upward and sideways.

Points on the pencil under water therefore appear raised, while the part above the water is seen in its true position. The two parts no longer join smoothly, and the pencil looks bent or broken at the surface. For the same reason a coin at the bottom of a pool appears raised, the pool looks shallower than it is, and letters under a glass slab appear displaced. If we look straight down from above (along the normal), the light does not bend and the pencil looks normal.

14.2 Why can a prism separate white light into its component colours while a plane glass slab cannot, even though both cause refraction?

Separation of white light into colours is called dispersion. It happens because the refractive index of glass is different for light of different colours: violet light is slowed down and bent most (largest n) and red light least (smallest n).

In a prism the light passes through two surfaces that are inclined to each other at an angle — the refracting angle of the prism. At the first surface the colours are bent by different amounts, and at the second surface the difference is increased further, so the colours emerge along clearly different directions and a band of seven colours (spectrum) is seen on the screen. The two refractions add together.

In a plane glass slab the two surfaces are parallel. Whatever separation the colours get at the first surface is exactly cancelled at the second surface, because the light is refracted back by the same angle. The emergent rays are parallel to the incident ray and the colours emerge together as white light again, only shifted sideways. So only a prism can disperse light and show a spectrum.

studysapiens.com

14.3 Why does light undergo total internal reflection when travelling from water to air at a certain angle but not the other way around?

Total internal reflection occurs only when light travels from an optically denser medium into a rarer one, and only when the angle of incidence is greater than the critical angle.

When light goes from water (denser, n = 1.33) into air (rarer, n = 1.00), it speeds up and bends away from the normal, so the angle of refraction is greater than the angle of incidence. As the angle of incidence is increased, the refracted ray bends more and more away from the normal, until at the critical angle (about 48.8° for water) the refracted ray grazes along the surface. If the angle of incidence is made larger than this, refraction is no longer possible and the whole of the light is reflected back into the water — total internal reflection.

In the reverse direction, from air into water, the light bends towards the normal and the angle of refraction is always smaller than the angle of incidence. The refracted ray can never reach 90°, whatever the angle of incidence, so there is no critical angle and total internal reflection cannot take place. This is why total internal reflection is a one-way effect, and why it happens at the water–air boundary only for light coming out of the water.

14.4 Optical fibres use total internal reflection for signal transmission. What would happen if the cladding had a higher refractive index than the core, and why?

If the cladding had a higher refractive index than the core, the core would become the rarer medium and the cladding the denser one. Then light travelling in the core would be going from a rarer to a denser medium, so it would bend towards the normal at the boundary and pass out into the cladding instead of being reflected back.

Total internal reflection requires the light to travel from the denser medium to the rarer medium at an angle greater than the critical angle. With the cladding denser, this condition can never be satisfied at the core–cladding boundary, so no total internal reflection would occur. The light would leak out of the core at every bounce.

The result would be a very large loss of signal: the light would escape through the sides of the fibre, the signal would become weak within a short distance, and the fibre would not work for communication. That is why the core of an optical fibre is always made of a material of higher refractive index than the cladding.

14.5 A person uses a magnifying glass to read small print. What must be the relationship between the object distance and the focal length of the lens for the image to be upright and magnified?

The object (the small print) must be placed at a distance less than the focal length of the lens, that is, between the lens and its principal focus (object distance < f).

In this position the rays coming from the object, after refraction through the convex lens, do not actually meet; they diverge as if they came from a point on the same side as the object. The eye sees a virtual image there. Because the image is on the same side, it is erect (upright), and because the rays are spread out more than the

studysapiens.com

object's own rays, the image is magnified.

If the object were exactly at the focus, the emergent rays would be parallel and the image would be at infinity; if it were beyond f, the image would become real and inverted. So for a simple magnifier the object must lie within the focal length, and the shorter the focal length, the greater the magnification.

D. Comprehensive Questions

14.1 Explain the terms normal, angle of incidence and angle of reflection in the context of light reflection. Provide a diagram to illustrate these terms.

Normal. The normal is the line drawn perpendicular to the reflecting surface (or to the boundary between two media) at the point where the incident ray strikes it. All angles in reflection and refraction are measured from this line, not from the surface.

Angle of incidence (i). It is the angle between the incident ray and the normal at the point of incidence.

Angle of reflection (r). It is the angle between the reflected ray and the same normal, measured on the other side of it.

Diagram. Draw a horizontal line for the mirror, mark the point of incidence O, and draw a dotted vertical line N–N′ through O as the normal. Draw the incident ray AO coming down at angle i to the normal, and the reflected ray OB going away at angle r to the normal on the other side. Label i between AO and ON, and r between OB and ON.

The law of reflection then states that i = r; the incident ray, the reflected ray and the normal all lie in the same plane, on the same side of the surface. The image formed by a plane mirror is virtual, erect, of the same size as the object and as far behind the mirror as the object is in front of it.

14.2 Define the terms critical angle and total internal reflection. Explain the conditions required for total internal reflection to occur.

Critical angle. When light passes from a denser medium into a rarer one, the angle of incidence for which the angle of refraction becomes exactly 90° (the refracted ray grazing the surface) is called the critical angle C of that pair of media. For water–air it is about 48.8°, for glass–air about 42° and for diamond–air about 24.4°.

Total internal reflection. When the angle of incidence in the denser medium is made greater than the critical angle, no light is refracted into the rarer medium at all; the whole of it is reflected back into the denser medium. This complete reflection inside the medium is called total internal reflection.

Conditions required: (i) the light must travel from an optically denser medium to an optically rarer medium (for example glass to air, or core to cladding in a fibre); (ii) the angle of incidence in the denser medium must be greater than the critical angle for that pair of media. If either condition is not met, ordinary reflection and refraction take place instead.

Applications: optical fibres for telephone and internet signals, endoscopes, totally reflecting prisms in periscopes and binoculars, brilliance of a diamond, mirage and the shining of a crack in glass.

studysapiens.com

14.3 Derive the equation (n = 1 / sin C) for the refractive index in terms of the critical angle C. Include all steps and assumptions.

Consider light inside a denser medium of refractive index n, striking the boundary with air (refractive index taken as 1). Let the angle of incidence in the denser medium be exactly the critical angle C. Then the refracted ray passes along the boundary, so the angle of refraction in air is 90°.

Step 1: write Snell's law for the boundary, taking the refractive index of the denser medium with respect to air as n:

$$n \sin i = 1 \times \sin r$$

Step 2: put i = C and r = 90°, because at the critical angle the refracted ray grazes the surface:

$$n \sin C = \sin 90°$$

Step 3: use sin 90° = 1:

$$n \sin C = 1$$

Step 4: rearrange for n:

$$n = 1 / \sin C$$

Assumptions: the second medium is air (or vacuum) of refractive index 1; the boundary is plane and smooth; the light is monochromatic, because the critical angle is slightly different for different colours. The same reasoning for two media of indices n_1 and n_2 gives $\sin C = n_2 / n_1$.

14.4 Describe the structure of an optical fibre. Write its any two advantages.

Structure. An optical fibre is a long, thin, flexible thread of very pure glass or plastic, usually thinner than a human hair. It has three parts:

1. Core — the innermost cylinder of glass of high refractive index (about 1.5), through which the light actually travels.

2. Cladding — a layer of glass or plastic of slightly lower refractive index surrounding the core; it keeps the light inside the core by total internal reflection and also protects the core from scratches.

3. Buffer coating / jacket — an outer plastic covering, and often a further protective sheath bundling many fibres together, which protects the fibre from moisture and mechanical damage.

Advantages: (i) very low signal loss, so a signal can travel tens of kilometres without amplifiers; (ii) enormous bandwidth — a single fibre can carry thousands of telephone calls and huge amounts of internet data at once; (iii) they are immune to electrical interference and to lightning, and cannot cause sparks, so they are safe in explosive places; (iv) they are thin, light and cheap to install compared with copper cables, and they cannot easily be tapped, so the transmission is secure.

studysapiens.com

14.5 What is long-sightedness? Discuss its causes.

Long-sightedness (hypermetropia) is a defect of vision in which a person can see distant objects clearly but nearby objects appear blurred. The near point of such an eye is farther away than the normal 25 cm.

Cause: the eyeball is too short (from front to back), or the eye lens is too weak, or the ciliary muscles cannot make the lens thick enough. As a result, the rays from a nearby object are brought to a focus behind the retina instead of on it, so the image on the retina is blurred. With age the lens also becomes less flexible — this form is called presbyopia.

Remedy: a convex (converging) lens of suitable positive power is used in spectacles. It converges the rays a little before they enter the eye, so the eye's own lens can bring them to a focus exactly on the retina. The power needed is found from the person's near point, using P = 1/f.

E. Numerical Problems

Sign rule used below: distances measured in front of the mirror (or on the object side of the lens) are taken positive and those behind it negative; the focal length is positive for a concave mirror and for a convex lens, and negative for a convex mirror and a concave lens. The formula used is 1/f = 1/v + 1/u, with magnification m = −v/u.

14.1 The speed of light in water is 2.25×10^8 m s^{-1}. Calculate the refractive index of water.

Given: $v = 2.25 \times 10^8$ m s^{-1}
$c = 3 \times 10^8$ m s^{-1}

To find: n = ?

Formula: n = c / v

Solution: $n = (3 \times 10^8) / (2.25 \times 10^8)$
n = 1.33

Answer: n = 1.33

14.2 A light ray in air ($n_1 = 1.0$) strikes a glass surface ($n_2 = 1.5$) at an angle of incidence 30°. Find the angle of refraction inside the glass.

Given: i = 30°
$n_1 = 1.0$
$n_2 = 1.5$

To find: r = ?

Formula: $n_1 \sin i = n_2 \sin r$

Solution: $\sin r = (n_1 \sin i) / n_2$
$\sin r = (1.0 \times \sin 30°) / 1.5$
$\sin r = (1.0 \times 0.5) / 1.5 = 0.3333$
$r = \sin^{-1}(0.3333)$
r = 19.47°

Answer: r ≈ 19.5°

studysapiens.com

14.3 A beam of light in air hits a glass slab at an incidence angle of 40°. Find the angle of refraction inside the glass.

Given: $i = 40°$
$n_1 = 1.0$
$n_2 = 1.5$

To find: $r = ?$

Formula: $n_1 \sin i = n_2 \sin r$

Solution: $\sin r = (1.0 \times \sin 40°) / 1.5$
$\sin r = 0.6428 / 1.5 = 0.4285$
$r = \sin^{-1}(0.4285)$
$r = 25.4°$

Answer: $r \approx 25°$ (25.4°)

14.4 A ray of light travels from air into an unknown liquid. The angle of incidence in air is 45°, and the angle of refraction in the liquid is 30°. Calculate the refractive index of the liquid.

Given: $i = 45°$
$r = 30°$
$n_1 = 1.0$

To find: $n_2 = ?$

Formula: $n_2 = (n_1 \sin i) / \sin r$

Solution: $n_2 = (1.0 \times \sin 45°) / \sin 30°$
$n_2 = 0.7071 / 0.5$
$n_2 = 1.414$

Answer: $n \approx 1.41$

14.5 A ray of light passes from air into a liquid. The angle of incidence in air is 50°, and the angle of refraction inside the liquid is 32°. The speed of sound in air is 340 m s^{-1} and in the same liquid is 1500 m s^{-1}. Calculate (a) the refractive index of the liquid (b) the speed of light in the liquid.

Given: $i = 50°$, $r = 32°$
$n_1 = 1.0$
$c = 3 \times 10^8$ m s^{-1}

To find: (a) $n = ?$, (b) $v = ?$

Formula: $n = \sin i / \sin r$, $v = c / n$

Solution: (a) $n = \sin 50° / \sin 32°$
$n = 0.7660 / 0.5299 = 1.446$
$n \approx 1.45$
(b) $v = c / n = (3 \times 10^8) / 1.45$
$v = 2.07 \times 10^8$ m s^{-1}

Answer: (a) $n = 1.45$ (b) $v = 2.08 \times 10^8$ m s^{-1}

studysapiens.com

14.6 Light travels inside a glass block of refractive index n = 1.6. It strikes the glass–air boundary. Calculate the critical angle for total internal reflection.

Given: n = 1.6
second medium = air

To find: C = ?

Formula: $\sin C = 1 / n$

Solution:
$\sin C = 1 / 1.6$
$\sin C = 0.625$
$C = \sin^{-1}(0.625)$
$C = 38.68°$

Answer: C = 38.68° (about 39°)

14.7 An object is placed 15.0 cm in front of a convex mirror, forming a virtual image 7.5 cm behind the mirror. What is the focal length of the mirror?

Given: u = +15.0 cm (object in front)
v = −7.5 cm (image behind the mirror, virtual)

To find: f = ?

Formula: $1/f = 1/v + 1/u$

Solution:
$1/f = 1/(-7.5) + 1/(15)$
$1/f = -0.1333 + 0.0667$
$1/f = -0.0667$
f = −15 cm

Answer: f = −15 cm (negative, so the focus is behind the mirror)

14.8 An object 30 cm tall is located 10.5 cm from a concave mirror with focal length 16 cm. (a) Where is the image located? (b) How high is it?

Given: u = +10.5 cm
f = +16 cm
h_o = 30 cm

To find: (a) v = ? (b) h_i = ?

Formula: $1/f = 1/v + 1/u$, $m = -v/u = h_i/h_o$

Solution:
$1/v = 1/f - 1/u$
$1/v = 1/16 - 1/10.5 = 0.0625 - 0.09524$
$1/v = -0.03274$
v = −30.6 cm
(b) $m = -(-30.6)/10.5 = +2.91$
$h_i = m \times h_o = 2.91 \times 30 = 87.3$ cm

Answer: (a) v = −30.6 cm, i.e. 30.6 cm behind the mirror (virtual, erect) (b) h = 87.3 cm

studysapiens.com

14.9 An object is placed at 30 cm to the left of a concave lens with focal length −15 cm. Find the image distance and describe the image.

Given: u = +30 cm
f = −15 cm (concave lens)

To find: v = ?

Formula: $1/f = 1/v + 1/u \Rightarrow 1/v = 1/f - 1/u$

Solution:
$1/v = 1/(-15) - 1/30$
$1/v = -0.0667 - 0.0333 = -0.1000$
$v = -10$ cm
$m = -v/u = 10/30 = +0.33$

Answer: v = 10 cm on the same side as the object — virtual, erect and one-third the size

The book prints −30 cm for this problem. That value comes from adding the two terms instead of subtracting them ($1/v = 1/f + 1/u$); with the formula used in the rest of the exercise the image distance is 10 cm. A concave lens always gives a virtual, erect and diminished image, and 10 cm agrees with that.

Formulae to Remember

- Snell's law: $n = \sin i / \sin r$, $n = c / v$
- Critical angle: $\sin C = 1 / n$, $n = 1 / \sin C$
- Mirror / lens formula: $1/f = 1/v + 1/u$
- Magnification: $m = -v/u = h_i/h_o$
- Power of a lens: $P = 1 / f$ (in metres), unit dioptre (D)

studysapiens.com